

From Governance Reviews to Task Substitution

DGF-Bench and the Role of Forward Deployed Engineers

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Abstract

Enterprise governance reviews transform project evidence into decisions, conditions, and actions. We evaluate whether tool-using language models can perform this work under explicit policies, using DGF-Bench: 300 synthetic Buy, Integrate, and Build projects with generated documents, architectures, simulated records, and sequential review gates. Agents have access to authoritative structured facts as well as documents. Across 899 evaluable model-project runs, Gemini 3.8 Flash, GPT-5.6 Luna, and DeepSeek v4.1 Flash achieve strict gate success of 94.98%, 83.29%, and 74.18%, respectively. Complete-route success is 76.92%, 42.33%, and 24.67%; recorded API cost totals USD 87.02. Gemini has 100% disposition accuracy: all its strict failures concern evidence conformity. We therefore distinguish decision quality from lexical traceability instead of interpreting the gate-route gap as a general reasoning failure. A new deterministic control executes the supplied public rules on the same structured facts and attains 1,700/1,700 strict gate successes and 300/300 complete routes without model calls. A post-hoc audit separates Gemini’s 85 evidence failures into 69 gates with structurally matching excerpts, nine with flattened cross-object excerpts, and seven lacking a required tool read. Thus this scaffolded task demonstrates review automation but no incremental LLM advantage. We formulate a conditional task-substitution hypothesis and specify how forward deployed engineers could connect review automation to operational evidence, authority, exception handling, and measurement of human work. A compact labor account shows why benchmark accuracy is not a staffing-reduction estimate. Independent human comparison, semantic evidence adjudication, scaffold-removal tests, and a prospective enterprise cohort are specified as follow-up studies, not completed experiments. A prospective seven-year hypothesis of 80% lower required FTE is specified relative to a future registered baseline; no enterprise cohort is yet enrolled. All benchmark sources, traces, and datasets are public. The new control and structural audit use existing dossiers and traces; no new model trials are reported in this revision.

Keywords: governance reviews; AI agents; forward deployed engineering; benchmark; task substitution; human workload.

1 Introduction: a gate is a contract, not a profession

A proposed enterprise project is examined by several groups before it proceeds. A legal reviewer checks contractual prerequisites; an architecture reviewer checks the technical design; security and readiness reviewers check protections and operational evidence. Each review consumes information and produces a decision with reasons, conditions, and obligations. The institutional label of the reviewer does not determine whether this transformation must always be performed by a person. The useful unit of analysis is the review contract: the inputs, rule, output, evidence, and authority that make a particular review acceptable.

This observation motivates a task-substitution hypothesis. If software can satisfy a review contract and its operational assurance requirements, it can replace some of the human execution of that

review. If it does so across a connected process while reducing total human work, the same volume of governed projects may require fewer staff. Neither implication requires abolishing the gate or lowering its standard. Nor does successful execution of a synthetic contract establish that every enterprise contract has already been captured.

The empirical question in this paper is specific: how reliably do three model endpoints apply explicit governance rules to synthetic dossiers, investigate with tools, produce evidence-bearing decisions, and continue through a sequence of gates? The experiment provides a useful early measurement because it exposes what an agent must return and retains what it actually read and did. It is also deliberately scaffolded: decision-relevant structured facts and policy rules are available to the model. Success is evidence about this policy-application task, rather than a direct comparison with unaided professional judgment.

1.1 Contributions and claim structure

The paper has three contributions. First, it documents a released benchmark with 300 distinct projects, 899 evaluable model–project runs, component-level scores, paired comparisons, and recorded costs. Second, it separates decision agreement, strict evidence conformity, and whole-route success, including concrete examples where their interpretations differ. Third, it defines an implementation and measurement agenda linking task automation to human work: the engineering artifacts an FDE must build, a minimal labor account, and follow-up studies that could substantiate or reject the workforce hypothesis.

Table 1: Three different claim levels. A stronger claim needs an additional measurement.

Level	Claim in this paper	Evidence required
Measured performance	Models often satisfy specified synthetic review contracts.	Released submissions, traces, scores, and uncertainty conditional on the dataset.
Deployment hypothesis	Comparable review tasks can be executed with less total human work.	Independent review quality and measured human hours in operational workflows.
Workforce projection	A registered cohort may test an 80% FTE reduction over a seven-year horizon.	A defined cohort, credible baseline, all-in labor boundary, and endpoint observations.

The second and third levels are research hypotheses. We do not call eventual complete automation a law: without a deadline or a specified comparison population, that endpoint cannot be rejected by a finite observation period. The useful empirical target is a change in required human work over a stated period with quality preserved. A substantial reduction can occur while some human tasks remain necessary.

1.2 Relation to prior research and to the companion manuscript

DGF-Bench belongs to a family of agent evaluations with different task and information contracts. WorkArena evaluates interactions with enterprise software (Drouin et al., 2024); ITBench evaluates operational IT scenarios across SRE, compliance/security, and FinOps (Jha et al., 2025). τ -bench combines policy-following, domain tools, simulated users, and evaluation of final database state, and explicitly measures reliability across repeated trials (Yao et al., 2024). CI-Work examines whether enterprise agents preserve contextual integrity while conveying useful information (Fu et al., 2026). AgentDojo evaluates tool use over untrusted data and prompt injection (Debenedetti et al., 2024).

The agentic BPM research agenda concerns process-level integration of agents, including planning, monitoring, and escalation (Calvanese et al., 2026). This paper contributes a small, inspectable review-workflow measurement within that agenda, not evidence that an entire business process has been automated. Its distinctive feature is the connected review contract and auditable artifact

Table 2: Task distinctions relevant to DGF-Bench; headline success percentages across these benchmarks are not directly comparable.

Benchmark	Task and evaluation object	Distinction from this DGF run
WorkArena	Enterprise web interactions and task completion	DGF uses a review/tool interface rather than browser interaction.
ITBench	Operational IT scenarios in SRE, CISO, and FinOps	DGF scores proposed reviews; it does not demonstrate operational remediation.
τ -bench	Policy-guided user/tool interactions; goal database state and repeated reliability	DGF has no live user conversation and its original run has one trajectory per pair.
CI-Work	Information flows, useful content, and privacy/contextual-integrity violations	DGF’s present score does not measure confidentiality under dense retrieval.
AgentDojo	Task execution and adversarial prompt-injection outcomes	DGF’s original cases do not provide an adversarial robustness test.
DGF-Bench	Connected gate decisions, findings, actions, lexical provenance, and mandates	Executable policy and authoritative structured facts strongly scaffold the task.

collection. The deterministic control reported below shows why high model scores on this particular contract should not be treated as evidence of uniquely linguistic capability.

The workload interpretation follows a task-level distinction between automation and aggregate employment: displacement of particular tasks can coexist with new tasks and changing demand (Acemoglu and Restrepo, 2019). Bainbridge’s analysis of automation motivates explicit attention to the work left for human operators (Bainbridge, 1983). Field evidence such as Brynjolfsson et al. (2025) illustrates a different empirical object: productivity observed in an operating organization. We do not transfer a productivity effect from another occupation to DGF reviewers or infer employment effects from model scores.

This is a shorter, empirical companion to *The Last Human Gate: Forward Deployed Engineering and the Automation of Enterprise Governance*, by the same author. Both manuscripts use the same September 2026 model experiment; they are not independent replications. This revision adds a deterministic control and a structural audit of saved evidence, with no new model trials yet. This paper removes the universal-law framing, reduces the scenario apparatus, and makes the validation gap an operational research plan. The original manuscript and data remain available separately.

The routes are author-designed synthetic workflows informed by practitioner experience. Their definitions, generated artifacts, rule implementation, and scoring interface are public, so the benchmark specification can be inspected without access to an unpublished practitioner document. This makes the implemented task inspectable; it does not establish that its rules represent a particular enterprise or an independently validated professional standard.

2 DGF-Bench: measured review performance on 300 synthetic projects

2.1 Question, population, and information available to the agents

This experiment asks whether an AI agent can apply a declared governance policy, identify the required findings and actions, cite observed evidence, respect its authority, and carry a review through a connected route. It measures review decisions and proposed actions, not implementation of remediation or changes in human workload. The field-validation studies proposed later in this paper have not been conducted.

Three practical questions organize the analysis: (1) does the model reach the disposition required by the rules, (2) does it produce all the supporting work needed for a valid review, and (3) does that reliability survive a sequence of reviews at an acceptable API cost? The unit called a *case* or

dossier is one generated project; a *gate* is one checkpoint on its route. Three models reviewing the same 300 projects produces 900 planned model–case runs, not 900 independent business projects.

The dataset contains 300 fictional projects: 100 Buy, 100 Integrate, and 100 Build cases. Buy and Integrate contain six gate occurrences each, and Build contains five, for 1,700 occurrences per model and 5,100 planned observations across three models. Cases were generated before these evaluations at difficulty setting 4, with sampling conditioned to improve decision coverage. Difficulty 4 is a generator configuration, not a validated scale of cognitive difficulty. For example, before decision-balancing rejection sampling, the architecture IP-overlap draw has probability $0.08 + 0.03 \times 4 = 0.20$ and the shared-service-principal draw has probability $0.10 + 0.03 \times 4 = 0.22$. Supplier generation permits sanctions hits and up to two failed mandatory criteria, and may select a non-optimal offer. Subsequent balanced sampling changes accepted outcome frequencies; these generator probabilities are not observed failure rates. The executable frozen source defines the setting, and no difficulty-level ablation is reported.

Project context, owners, budgets, data classifications, technical architectures, documents, and simulated system records derive from canonical case facts. Documents can be missing, stale, or inconsistent. Specialist reviews precede a final General review; the routes are examples rather than a universal enterprise standard.

The agent receives gate objectives, a structured output contract, an explicit decision policy, and tools to read allowed evidence and query the synthetic environment. In particular, the policy includes executable rule definitions and identifies an authoritative structured `REVIEW_FACTS` snapshot. The evaluator-only reference file is inaccessible through the agent’s tools. This is nevertheless a strongly scaffolded test of policy application: the agent need not discover the rules or reconstruct every decision-relevant fact from unstructured documents. Architecture images are available under the vision setting; image availability is not evidence that image interpretation was necessary for success. These design choices limit claims about expert reasoning on independently collected enterprise dossiers.

The dataset audit reports 300 distinct project identifiers, architecture signatures, and complete decision-fact signatures, but only 267 distinct whole-case finding patterns. Gate-specific relevant fact signatures include 142 of 200 Compliance cases and 241 of 300 Security cases. All rules applicable to the selected phases are covered according to the generator’s audit. Phase-adjusted normalized decision entropy ranges from 0.9971 to 1.0000. Balanced labels improve coverage, but do not estimate enterprise prevalence or establish independent business validity.

2.2 What is generated, and what the model actually does

The case generator is a program using seeded facts, rules, and document templates. The AI model under evaluation is the reviewer, not the author of the fictional need or architecture. A Buy scenario represents acquiring a solution; Integrate represents connecting existing systems; Build represents developing a new project. Those scenarios are given budgets, users, owners, dates, data classifications, and technical constraints. A generated name is a scenario label, not an independently elicited customer requirement.

1. **Construct the project facts.** The program fixes what is true in the synthetic case, such as the requested budget, whether a supplier has supplied a document, or whether a restore test succeeded. Decision-coverage sampling produces cases with different required outcomes.
2. **Generate the design and dossier.** The same case drives an architecture diagram, project charter, gate review requests, and evidence files. A reviewer may receive a technical design, network/data-flow tables, contract records, security evidence, or readiness tests, depending on the gate (Table 3). Some presented evidence is incomplete or stale.
3. **Start a gate review.** The harness supplies the project context, gate and phase, applicable

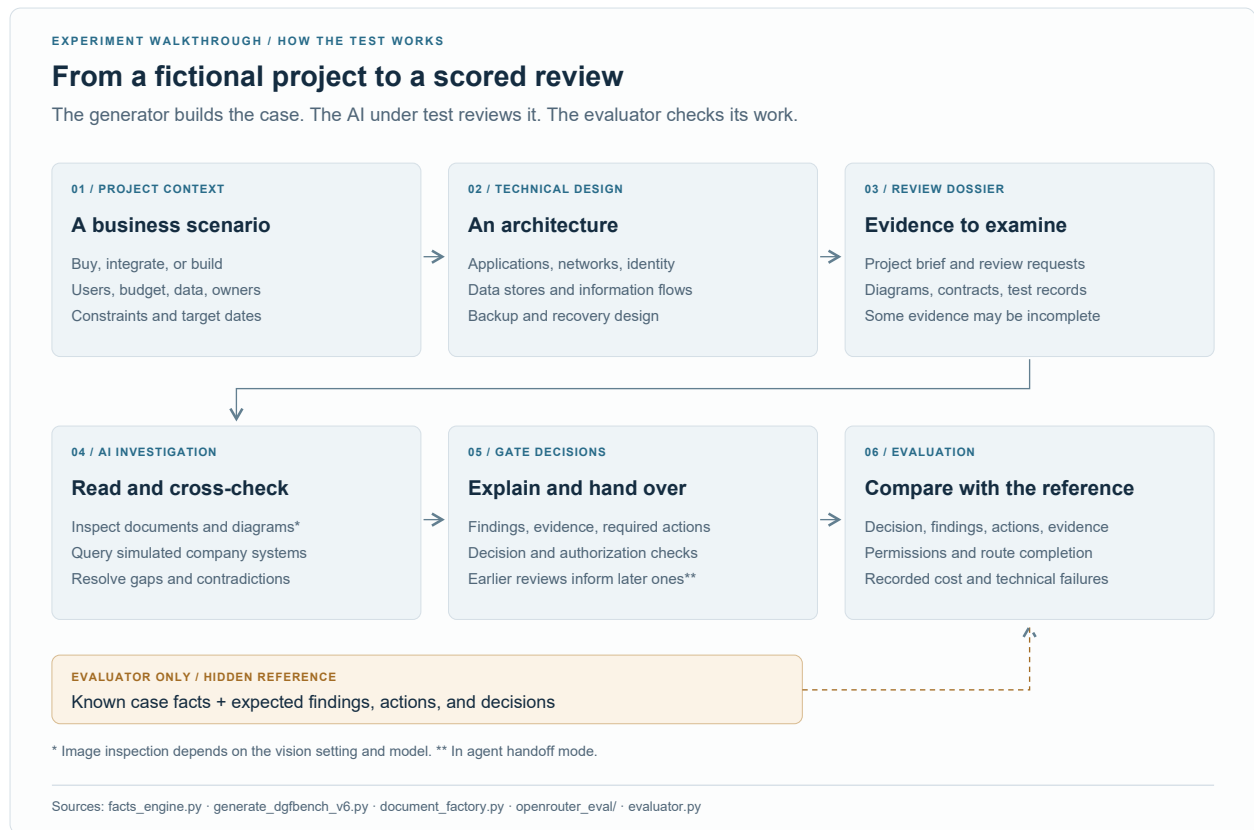


Figure 1: Experimental workflow. The generator constructs the project and evidence; the tested model investigates and submits reviews; the evaluator uses a separate reference. The diagram describes the implemented test, not a measured enterprise deployment.

policy, permitted evidence and tools, and the answer schema. The model decides which allowed records to inspect. It can list and read evidence, query simulated systems, request information, and inspect existing authorization mandates.

4. **Submit a decision.** The model returns a disposition, finding identifiers, proposed actions, evidence references and supporting excerpts, an authorization indicator, and a rationale. A typed submission is recorded. Producing a syntactically valid record alone does not satisfy the substantive score.
5. **Continue the route.** In the tested **agent** mode, the harness passes actual upstream model reviews to later checkpoints. The final General review consolidates specialist opinions and its own checks. The tested routes are listed in Table 4.
6. **Evaluate and retain the trace.** A deterministic evaluator compares the submitted record and observed tool events with the reference. It records component scores, full gate and route success, critical misses, false approvals, provider failures, and usage. It is not another LLM judge. Failed or incorrect decisions are retained; infrastructure exclusions are reported.

The five dispositions are **GO** (proceed), **GO_WITH_RESERVATIONS** (proceed under conditions), **REWORK** (revise the proposal), **SUSPENSION** (wait for a prerequisite), and **NO_GO** (do not proceed as submitted). Approval is therefore not the target outcome for every dossier: a justified refusal is a successful review. Tool actions take place in the synthetic environment; a request to fix a defect is not evidence that the defect was fixed.

Table 3: Concrete examples of generated review material. The contents vary with the route and gate. Authoritative structured snapshots are available in addition to these artifacts.

Material	Examples in the case	Review question
Project context	Charter and review request (Word), project context (JSON)	What is proposed, for whom, at what cost, and under whose ownership?
Architecture	Diagram (SVG/PNG), technical design, network and flow tables	Do the design facts meet the declared constraints?
Security	Identity, vulnerability, monitoring, and resource records	Are the protections required by the policy present?
Supplier and legal	Vendor evidence, contract versions, data-processing agreement status	Are the supplier and contractual prerequisites satisfied?
Readiness	Backup, restore/failover and load-test records; runbook	Can the proposal demonstrate the required operational readiness?

Table 4: The three routes actually evaluated; each project ends with General. These are the complete route sequences evaluated in this study.

Route	Cases	Ordered gates
Buy	100	Procurement, Legal, Compliance, Security, IT, General
Integrate	100	IT, Architecture, Security, Legal, Compliance, General
Build	100	IT, Architecture, Security, Tech Readiness, General

2.3 A generated project and a recorded successful review

Case DGF-BLD-035200_build, *Project Falcon—Secure Vendor Portal*, provides a concrete example. Its generated HR context has 1,000 users, a requested budget of EUR 500,000, confidential data, high business criticality, and a primary region of UAE North. Figure 2 reproduces its actual evidence diagram. It depicts identity, application and data services, monitoring and backup, and a secondary recovery region. Its labels are claims in the dossier to be checked, not an independently validated production blueprint.

In Gemini’s recorded Tech Readiness review, the authoritative record has a draft runbook. The agent identifies TR-OPS-001 and proposes completing the operational handover. Other required readiness checks pass. It cites the observed excerpt `"runbook_status": "draft"`, consults the active mandate, and executes the simulated conditional-approval tool with a condition to approve the runbook before production cutover. The tool reports `EXECUTED` and explicitly leaves the finding unresolved. The submitted outcome is therefore `GO_WITH_RESERVATIONS`, with the finding and required action retained and the authorization indicator set to true.

All three models pass all five gates of this particular dossier. Their observed dispositions are `GO`, `GO`, `GO`, `GO_WITH_RESERVATIONS`, and `GO` in route order. This example was selected after evaluation to explain the mechanics, not as a representative sample or an additional experiment. The study-level counts include every evaluable case, including unsuccessful reviews. The public research directory includes `worked_examples.json`, with the original archive-relative paths and selected recorded outputs for this example and the evidence-failure example below. The complete traces remain in the release.

2.4 Protocol, completion, and outcomes

Table 5 identifies the three tested endpoints. Each receives the same case population, policy, tool interface, and task limits; no model is trained or fine-tuned by this experiment. The comparison concerns these deployed endpoints under the recorded conditions, not different implementations of a human review process. Configuration was temperature 0, vision `auto`, agent-produced handoffs, at most 20 turns and 40 tool calls per gate, and 8,192 output tokens per turn. Gates remained

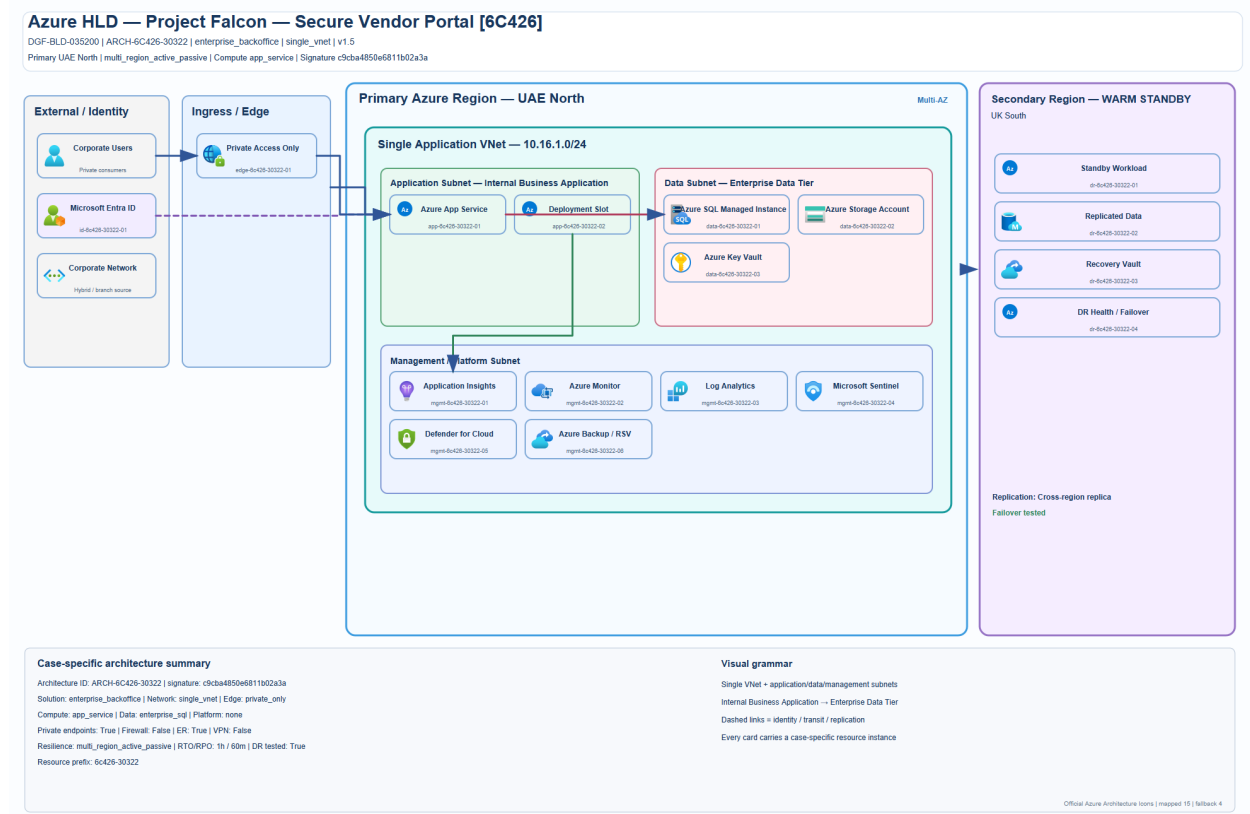


Figure 2: An actual generated architecture document from Project Falcon. Full-resolution evidence is in the released dataset. The figure illustrates what a dossier can contain; it does not imply that this image was necessary to obtain the correct review outcome.

sequential within a project. The last resume used 12 workers, four per model, and a USD 110 cost cap; an earlier invocation used six workers and a USD 65 cap. Data collection spanned 22–23 September 2026. Resumption retained completed cases and valid gate checkpoints. Historical provider/key-limit failures remain in the released traces. The 78.73-minute elapsed time in the last run statistics describes that resume only.

Of 900 planned model-case runs, 899 are evaluable: 300 for DeepSeek, 299 for Gemini, and 300 for Luna. Gemini case DGF-INT-035128_integrate failed at its first IT gate with a provider generation error (`finish_reason=error`); it is excluded as an infrastructure failure. Thus 5,094 of 5,100 planned gates are evaluable, with all expected gates attempted in the evaluable cases. There are no recorded final agent-protocol failures or model-identifier mismatches. Provider-reported identity is not an independent audit of underlying weights. DeepSeek was served by multiple OpenRouter providers; Gemini and Luna were recorded through Google AI Studio and OpenAI.

Table 5: Models used in the experiment. Names are short labels for the exact requested OpenRouter identifiers, not claims about independently verified model weights.

Model label	Requested identifier	Recorded serving provider
DeepSeek v4.1 Flash	deepseek/deepseek-v4.1-flash	Multiple providers
Gemini 3.8 Flash	google/gemini-3.8-flash	Google AI Studio
GPT-5.6 Luna	openai/gpt-5.6-luna	OpenAI

Scoring uses DGF-decision-v8.2-authorized-review. A gate succeeds strictly only if its disposition, finding set, action set, evidence provenance, and authorization indicator all satisfy the

reference. Findings and actions require F1 equal to one. Evidence requires an observed reference and exact excerpts supporting the relevant field/value tokens. This is lexical provenance checking, not independent verification of arbitrary reasoning. Authorized conditional approvals can change the effective reference disposition; the evaluator verifies the mandate and recorded action. General combines effective reference opinions, so an upstream error can also cause a downstream failure. Complete-route success requires every gate to succeed strictly; it does not mean that remediation was executed.

For model m , let s_{mcg} be one when gate g of case c passes every component, and zero otherwise. The two principal rates are

$$\text{GateCSR}_m = \frac{\sum_c \sum_{g \in G_c} s_{mcg}}{\sum_c |G_c|}, \quad \text{RouteSuccess}_m = \frac{\sum_c \prod_{g \in G_c} s_{mcg}}{|C_m|},$$

where C_m contains the evaluable cases and G_c their expected gates. This gate rate is weighted by gate count: a six-gate route contributes six observations and a five-gate route contributes five. A model passing four gates of a five-gate case has 80% gate success for that case but zero complete-route success. Decision-only accuracy checks just the submitted disposition. A critical miss is an omitted reference finding classified as critical; a false approval is `GO` or `GO_WITH_RESERVATIONS` when the effective reference requires rework, suspension, or refusal.

Gate confidence intervals use a route-stratified bootstrap of whole cases, preserving dependence between a project’s gates (2,000 draws, seed 81931; a case-count Wilson fallback handles degenerate intervals). Route intervals use Wilson bounds. Pairwise differences use the same 299 cases for all models, resampled together within routes (10,000 draws, NumPy seed 81931). These are descriptive intervals conditional on the synthetic sampling design. No preregistered analysis or independently adjudicated ground truth is claimed, and one trajectory per model and case does not estimate run-to-run inference variability.

2.5 Overall and route-level results

Table 6: Measured synthetic benchmark outcomes. Gate CSR requires complete conformity; route success requires all gates. Cost includes recorded failed attempts.

Model	Cases	Gate CSR [95% CI]	Route success [95% CI]	USD
DeepSeek	300	74.18% [71.88, 76.47]	24.67% [20.13, 29.84]	10.09
Gemini	299	94.98% [93.86, 96.04]	76.92% [71.82, 81.34]	71.88
Luna	300	83.29% [81.29, 85.24]	42.33% [36.87, 47.99]	5.05

Gemini succeeds strictly on 1,609 of 1,694 gates, Luna on 1,416 of 1,700, and DeepSeek on 1,261 of 1,700. Corresponding complete-route counts are 230 of 299, 127 of 300, and 74 of 300 (Table 6). Mean weighted component scores, which are not strict success rates, are 99.68%, 97.67%, and 95.02%, respectively. The distinction matters: strong average components can coexist with many incomplete project reviews.

On the 299 common cases, CSR is 94.98% for Gemini, 83.23% for Luna, and 74.09% for DeepSeek. Paired differences are 20.90 percentage points for Gemini minus DeepSeek (95% interval 18.30–23.44), 11.75 for Gemini minus Luna (9.62–13.93), and 9.15 for Luna minus DeepSeek (6.67–11.63). The ranking persists on the matched cohort; this is not a universal model ranking.

Gemini’s route success falls from 87 of 100 Buy cases to 68 of 100 Build cases. Luna succeeds on 20 Buy, 52 Integrate, and 55 Build cases; DeepSeek succeeds on 13, 32, and 29, respectively (Table 7). These differences do not isolate the causal effect of route length, because gate composition and case content also differ.

Procurement is the lowest-CSR gate for both Luna (33%) and DeepSeek (43%). Luna’s General CSR is 61.67%. Luna exceeds Gemini’s descriptive CSR in Architecture and Tech Readiness (Table 8);

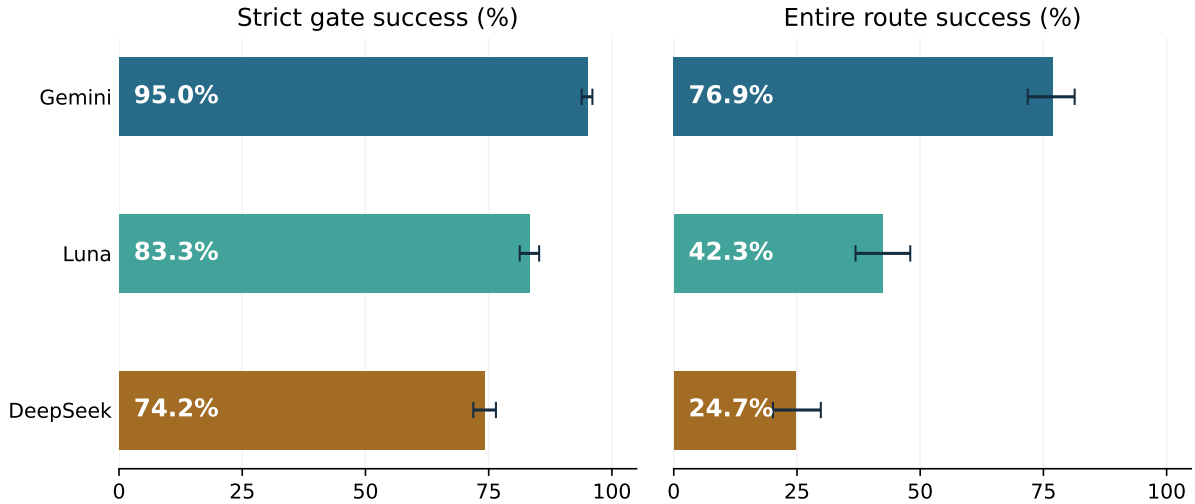


Figure 3: Measured gate and complete-route review outcomes. Intervals are computed over independent case units rather than treating all gates as independent trials. These are model observations on synthetic dossiers, separate from the workforce scenario figures.

Table 7: CSR and complete-route counts by project route. Gemini excludes one Integrate case.

Route	DeepSeek: CSR; routes	Gemini: CSR; routes	Luna: CSR; routes
Buy	67.83%; 13/100	97.67%; 87/100	74.67%; 20/100
Integrate	78.17%; 32/100	95.12%; 75/99	87.67%; 52/100
Build	77.00%; 29/100	91.60%; 68/100	88.40%; 55/100

Table 8: Strict success by gate family (percent). Counts vary by route composition; complete denominators and intervals are supplied in the released tables.

Gate	DeepSeek	Gemini	Luna
Architecture	75.00	85.43	89.50
Compliance	88.00	100.00	96.00
General	68.67	87.96	61.67
IT	87.00	100.00	97.67
Legal	68.50	99.50	80.50
Procurement	43.00	96.00	33.00
Security	72.33	98.66	92.00
Tech Readiness	71.00	89.00	97.00

these local comparisons are exploratory, without multiplicity-adjusted tests or a claim of specialist superiority.

2.6 Evidence failures, unsafe approvals, and cost

Disposition accuracy alone is 100% for Gemini, 98.12% for Luna, and 95.24% for DeepSeek. All 85 Gemini strict failures concern evidence conformity; findings, actions, dispositions, and authorization indicators match the reference. Evidence is the sole failing component in 221 of 284 Luna failures and 338 of 439 DeepSeek failures. Invalid excerpts and missing or unobserved references are grouped by these diagnostics; not all are harmless formatting defects. Changing the evidence rule after observing results would change the estimand. A semantic evidence metric would require a separately declared analysis.

A recorded example of a correct decision with a failed proof. In DGF-BLD-035201_build, Gemini reads the Architecture facts: an API gateway is required but absent, and measured latency is 55 ms against a 20 ms target. It correctly names ARCH-API-001 and ARCH-PERF-001, proposes adding the gateway and remediating latency, and submits REWORK. Two attempted conditional approvals are rejected by the tool: one omits an open finding and the other includes a finding ineligible for risk acceptance. The submitted refusal to proceed is consistent with those restrictions.

The gateway excerpt reverses two fields. It places `api_gateway_present` before `api_gateway_required`; the observed record has the reverse order. The values are correct but the excerpt is not an exact observed sequence. Evidence fidelity is 0.5 for the two findings, so strict success is zero while disposition, finding, action, and authorization scores are all one. The weighted component score is 0.95. This selected trace explains exactly how a fluent, substantively correct review can fail this benchmark’s evidence contract; it does not justify silently relaxing that contract after seeing the results.

DeepSeek records one critical finding miss and one false approval, both in the Architecture review of DGF-INT-035136_integrate. An address overlap required NO_GO; at forced finalization, the model submitted GO with an empty finding set, no evidence, and a “placeholder” rationale. The response passed the submission schema but failed the substantive score. This observation combines decision quality with the ability to finalize under resource limits.

Luna records three critical misses and one false approval. In case DGF-INT-035197_integrate, its Legal review approved a dossier with a missing DPA, while its own rationale said that REWORK was required. The other critical misses concern private endpoints in Security and an upstream NO_GO in General. The latter can include error propagation and should not be interpreted as an independent new safety event. Gemini has no observed critical miss or false approval in its evaluable runs; this does not establish zero risk. Recorded truncated-response counts are 2 for DeepSeek, 14 for Gemini, and 0 for Luna, without implying that these counts equal incomplete cases.

The usage ledgers sum to USD 87.015829172, agreeing with the cumulative budget counter. Gemini accounts for USD 71.88, DeepSeek USD 10.09, and Luna USD 5.05, including recorded retries and failures. Approximate cost per evaluated dossier is USD 0.240, 0.034, and 0.017, respectively. Luna therefore combines a higher overall strict score with lower recorded cost than DeepSeek in this execution. Costs depend on provider routing, caching, price, and retry history; they are not a labor-cost comparison or a permanent pricing claim.

3 What the scores establish

3.1 A near-saturated disposition task and a stricter trace contract

Gemini’s 100% disposition accuracy is a strong result under the declared conditions. It also indicates that this sample offers little remaining discrimination on that component for this endpoint. The model can read explicit rules and authoritative structured facts; this experiment does not isolate how much ability is needed to recover those facts from documents. The deterministic control now executed on the same 300 dossiers reaches full strict success. It demonstrates that the supplied executable rules and structured facts are sufficient to solve this instance of the task without model inference. Model success remains an observation about model-based execution, but does not establish an incremental benefit over direct rule execution.

Strict success answers a different question: does the entire submitted review satisfy the implemented contract, including exact evidence provenance? It is useful when downstream processing requires precise traceability, but it is not a direct measure of independent expert judgment. The selected Architecture example shows a correct decision and correct values with a failed excerpt because two fields were reversed. The Legal false approval by Luna and DeepSeek’s placeholder approval illustrate materially different errors. Those classes should not be collapsed into a single narrative

about reasoning failure.

Table 9: Separate observed outcomes. Evidence-only failures have no other failing scored component; they are not all established to be harmless formatting errors.

Outcome	DeepSeek	Gemini	Luna	Rules
Correct dispositions / evaluable gates	1,619/1,700	1,694/1,694	1,668/1,700	1,700/1,700
Strictly successful gates	1,261	1,609	1,416	1,700
Evidence-only gate failures	338	85	221	0
Other strict gate failures	101	0	63	0
False approvals	1	0	1	0
Critical finding misses	1	0	3	0

The strict gate–route gap for Gemini is consequently a compounding of evidence-conformity failures across routes, not evidence of incorrect final dispositions. A strict route fails when any component at any gate fails. This is the intended operational definition, but the figure 76.92% must be read with that definition. We retain both the original scoring rule and the component breakdown. Neither retrospectively relaxing the rule nor interpreting every lexical failure as a dangerous decision is justified.

3.2 Executed deterministic control

A separate generation process reads each dossier’s public route and gate contract. For each occurrence it reads the authoritative snapshot through the same public evidence reader, executes `rule_definition_python` from the supplied policy, combines finding dispositions using the declared priority order, and sets the authorization indicator using the declared disposition list. Finding construction and result aggregation are implemented from that public interface. The policy functions are the executable definitions supplied to the language models; they are not a separately learned policy or an independently validated governance standard.

The comparator retains the base disposition and never requests optional conditional risk acceptance, a strategy explicitly permitted by the policy. It cites complete observed JSON snapshots as exact evidence excerpts and propagates its own earlier decisions to General. This exploits the absence of a quote-length cap; the program is not subject to the models’ per-turn output-token limit. It does not attempt document extraction or image interpretation. A read guard blocks evaluator-only reference files during generation, and the generation process asserts that neither the evaluator nor scorer module was imported. All predictions and observation records are written before a second process invokes the frozen scorer, which is allowed to read the hidden references for evaluation.

The result is **1,700/1,700 strict gate successes and 300/300 complete routes**, with no false approvals or critical misses. Model API expenditure is zero; programming and local compute are not costed. On the 299 cases shared with Gemini, the control also succeeds on all 1,694 gates. Optional approval choices can differ between systems, and each is scored against the protocol’s effective reference; full conformity does not mean identical submitted decisions.

This is a scaffold-sufficiency control. It does not validate the business rules independently, because the public policy and evaluator share their definitions. It does show that an LLM is unnecessary for achieving the measured endpoint when those functions and structured facts are already available. The main unresolved model contribution therefore lies in tasks that this run did not isolate: recovering facts, resolving ambiguity, and interacting with less structured operational evidence. Claims about such a contribution require the proposed ablations and external validation, rather than a higher score on this saturated contract.

3.3 Executed structural audit of Gemini’s 85 failures

We inspected every Gemini strict-failure gate against its saved submission and pre-submission read events. The 85 failed gates contain 84 flagged finding-support items: a gate may contain several items, while a citation-only failure may contain none. The audit parses each quote as a JSON object fragment, rejects duplicate keys, and checks all quoted field values against a single observed object while ignoring key order and numeric serialization such as 30 versus 30.0. Booleans remain distinct from numbers; 10 cannot match 100. A second diagnostic records where values exist across different objects, without treating a flattened join as a faithful single-object excerpt. This is a post-hoc structural test, not independent expert adjudication.

Table 10: Exhaustive post-hoc audit of Gemini strict failures. Categories describe recorded representation and observation defects, not new semantic pass rates.

Gate category	Gates	Item-level explanation
All flagged quotes match observed objects after structural normalization	69	75 flagged finding items preserve the checked field values within an observed object.
Flattened cross-object excerpts	9	9 flagged items combine observed values from separate objects; paths are retained for inspection.
Required evidence-tool observation absent	7	All cite <code>UPSTREAM_DECISIONS</code> without the corresponding read event.

For example, a flattened high-availability excerpt combines business criticality from the project object with redundancy flags from the architecture-profile object. Those values are present, but the quoted combined object was not observed. In the seven citation cases, upstream reviews were also included in the gate prompt; the defect is the missing required evidence-tool read, not proof that the model never received the underlying information. Neither a corrected quote nor an invented read event is inserted into the original trace.

The audit identifies representation-related causes for the 78 excerpt-failure gates and a separate observation-contract cause for seven gates. It supplies no independent judgment that every policy inference is justified or that every necessary premise was cited. The single-object matching check accepts relevant field values rather than proving full entailment, as does the original lexical checker. We retain the original 94.98% strict gate and 76.92% strict route rates and report the audit alongside them. No revised whole-benchmark semantic rate is claimed. Complete excerpts, observed contents, matched paths, categories, and source trace identifiers are released with the audit script.

3.4 A proposed semantic evidence assessment

A semantic assessment should supplement, rather than replace, the frozen lexical score. Its object would be the relationship between an output claim and evidence the agent actually observed. A correct conclusion based on an unseen file would not pass an observed-evidence requirement. A correctly observed passage that does not support the conclusion would not pass semantic support. Harmless serialization changes should be distinguished from both.

We propose four separately recorded judgments for each required finding: whether the source was available and read before submission; whether the cited span is correctly identified; whether it entails the factual claim with its scope, date, and qualifications intact; and whether the claim is sufficient to support the relevant policy finding. Each receives a supported, unsupported, or ambiguous label and a written reason. Disputed policy interpretation is recorded separately from a disputed quote. Unsupported observations must not be repaired by showing the adjudicator the hidden reference answer as if the agent had seen it.

Two domain-qualified adjudicators would label the same items independently, with model names and original pass/fail outcomes masked where feasible. A third adjudicator would resolve disagree-

ments under a written rubric. Report raw agreement, a chance-corrected agreement statistic, the disagreement categories, and results both before and after adjudication. Model-generated judging can assist triage only after comparison with the human labels; it must not define its own validity.

For an initial audit, include all strict evidence failures and a probability sample of passing items stratified by model, route, and gate. Use the sampling probabilities for any population estimate. If the goal is a new complete-route semantic score, adjudicate every required evidence item on the routes used for that estimate; a convenience sample of failing excerpts is insufficient. Freeze the rubric on development examples and evaluate it on held-out items. Because the existing traces have already been inspected, this is a declared post-hoc audit, not a preregistered re-evaluation of the original primary endpoint.

This paper reports a structural audit but no independently adjudicated semantic pass rate. The practical value of the proposal is to make the next measurement answer a precise question: how often does strict trace failure indicate unsupported reasoning, and how often does it indicate a representation defect? The distinction matters for both reliability claims and engineering effort.

3.5 Dependence, uncertainty, and error propagation

The benchmark contains 300 underlying projects, not 5,094 independent enterprises. Multiple gates within a project share facts and handoffs; multiple models review the same project. The released case-cluster and paired analyses preserve those dependencies for the reported intervals. Those intervals still condition on the generated population and one retained trajectory per model and case. They do not measure uncertainty over organizations, model versions, or repeated inference runs. Temperature zero does not supply a replication study.

Sequential handoffs allow an upstream output to affect a later review. They make propagation observable in traces, but this single handoff condition does not identify a causal propagation effect. A controlled comparison would run the same cases with agent handoffs, reference handoffs, and no prior-review content, with matched resource budgets and repeated trajectories. Report downstream changes conditional on the upstream error class, rather than treating every repeated mention of one defect as an independent safety event.

The original gate definition is a useful strict endpoint; it is not the only endpoint an operating organization would choose. A deployment evaluation should separately register false approvals, missed critical findings, evidence validity, unauthorized actions, cycle time, and human work. Acceptance thresholds should follow the consequence of each error. The present study supplies observations for several of those outcomes, but no enterprise has validated its contract as a sufficient production acceptance test.

4 The role of forward deployed engineers

Here, a forward deployed engineer (FDE) denotes an engineer working with an organization’s domain owners to integrate and operate a review system. The contribution of the role is not assumed from its title. It consists of deliverables that make a review contract executable, observable, and maintainable. The benchmark already supplies many of these prerequisites; in an enterprise they must be built and kept correct.

4.1 From a generated contract to an operational service

The first task is policy formalization. A specialist and the policy owner must agree which decision conditions are mandatory, which permit an exception, and which require interpretation. The implementation should record version, scope, effective date, and responsible authority for each rule. A model’s explanation is not a substitute for resolving contradictory policies. Ambiguities

discovered during formalization are findings about the organization’s process and should remain visible in the deployment record.

The second task is evidence integration. Synthetic snapshots make facts convenient to read; real evidence comes from versioned repositories, identity services, ticketing systems, monitoring, contracts, and people. Each connector needs access control, freshness checks, stable source identifiers, and recorded observations. If two sources disagree, the system needs an escalation or precedence rule rather than silently selecting whichever supports approval. The cost of producing a reliable snapshot belongs in the deployment, even if the subsequent model review becomes almost trivial.

The third task is authority. Recommendation, conditional approval, and execution are distinct capabilities. The reviewer must act within a versioned mandate whose scope includes the project, action, and any conditions. An exception mechanism must retain the unresolved finding and identify who accepts it. Approval status must not be inferred solely from a fluent rationale. The benchmark’s rejected conditional-approval attempts make this distinction concrete: a proposed call can be invalid even when a later submitted decision is correct.

The fourth task is a shared case state. Later gates need to know which obligations remain open, which evidence is stale, and which prior opinions have been superseded. Passing an unstructured summary alone can lose conditions; copying every historical record can create contradictions. An implementation should retain typed decisions, evidence references, policy versions, unresolved obligations, and the causal relationship between reviews. Gates can be introduced incrementally while using the same state and interface design.

Table 11: Proposed FDE deliverables and independently checkable acceptance evidence. These are implementation requirements, not deployment outcomes measured in this benchmark.

Deliverable	Concrete artifact	Acceptance evidence
Policy contract	Versioned rules, exception scope, output schema	Specialist adjudication of held-out cases and documented disagreements.
Evidence interface	Connectors, timestamps, source IDs, read log	Tests of stale, missing, contradictory, and access-restricted records.
Authority boundary	Mandate registry and enforced action checks	Invalid calls rejected; accepted actions trace to an active mandate.
Connected case state	Findings, obligations, decisions, dependency graph	Replayed handoffs preserve open issues and invalidate stale approvals.
Review service	Agent, validator, escalation and recovery behavior	Held-out quality results including resource exhaustion and provider failure.
Operating account	Human activity ledger and support allocation	Measured effort includes specialists, engineers, suppliers, and rework.

4.2 Failure-informed engineering priorities

The observed failure classes suggest testable interventions. An evidence serializer could attach verbatim spans and stable source identifiers directly from the read log, avoiding a model’s reconstruction of exact quotations. A consistency check could flag a submitted approval whose rationale explicitly asks for rework. A finalization contract could require abstention or escalation when resource limits prevent a supported answer, instead of permitting a placeholder approval. A handoff validator could ensure that an unresolved upstream prohibition cannot silently disappear from the final review.

These interventions could reduce particular failures, but their effects are not measured here. Each should be evaluated as a new system version on held-out cases, recording any new failure modes and human burden. A validator that catches an error but routes every case to a specialist may improve decision safety without reducing labor. A serializer that makes quotes exact may raise strict CSR without improving substantive judgment. The outcome attributed to an intervention should match what it changes.

Operational performance is a property of the complete system: policy owners, evidence systems, model, validators, authorization controls, and exception handling. The FDE’s recurring work is part of that system. Initial integration hours are a transition investment; repeated policy changes, connector repairs, evaluations, incident investigations, and model migrations are recurring work. Moving them into an external supplier does not remove them from the DGF labor boundary. Separate reporting of build effort and recurring effort enables both an annual capacity estimate and a cumulative payback analysis.

4.3 Testing the implementation hypothesis

We hypothesize that a connected implementation with shared evidence and obligations can avoid some re-reading, re-entry, and reconciliation across review teams. The relevant comparison is with locally optimized gate tools at comparable project mix, authority, quality, and automation scope. Deployment can remain incremental in both conditions. Measure duplicated evidence collection, downstream correction, specialist review hours, recurring support, and total quality-constrained human work. Assign implementation approaches across comparable workflows where feasible, or use a staged comparison with explicit confounding controls.

An improvement would support the value of the shared implementation, not prove that a particular job title caused it. No FDE deployment is included in the current experiment. Likewise, the idea that governance will automate before other business functions is a comparative hypothesis: it requires matched functions, comparable outcome definitions, and observed deployment times. The present results do not establish a universal ordering. They identify a concrete candidate for further study because much of the review contract can be specified and inspected.

5 From task success to required human work

5.1 A minimal accounting identity

Automating a review can remove reading, comparison, documentation, and coordination work. It can also add exception handling, monitoring, correction, and maintenance. To determine the net effect, first define the same organizational boundary and project volume in both regimes. Let V be reviews per measurement period and h_0 the baseline mean human hours per review. Let p be the fraction routed to substantive human exception handling, h_E the total human hours on each such review, h_S the human hours on a standard-path review, and h_W additional rework averaged over all reviews. Let B_0 and B_1 denote case-independent support hours before and after deployment. Then

$$L_0 = Vh_0 + B_0, \tag{1}$$

$$L_1 = V\{ph_E + (1 - p)h_S + h_W\} + B_1, \quad N_j = L_j/K \quad (j = 0, 1). \tag{2}$$

Here K is a fixed useful-hours convention per FTE per period. The same convention is used in both regimes. Exception time includes any preliminary review of those same cases. Rework excludes activities already counted in exception time, ordinary review, or separately counted repeat visits. Support includes internal and external work within the registered function.

Equation (2) is accounting, not an estimated response to AI adoption. With consistent inputs it is equivalent to adding the underlying activity hours directly. We make no hypothesis that it outperforms an algebraically identical comparator. Its purpose is to expose omissions and make an empirical claim about savings auditable. Gate success, approval rate, and the share of tool calls completed are not substitutes for any of these human-time variables.

5.2 Why remaining cases can retain disproportionate work

Let w denote the exception group’s share of baseline case-handling hours, under a fixed partition of the same case population. Let a be its post-deployment effort divided by its own baseline effort,

and u the corresponding ratio for the standard group. Put $r = h_W/h_0$ and $b_j = B_j/(Vh_0)$. For positive baseline group means,

$$R := \frac{N_1}{N_0} = \frac{wa + (1-w)u + r + b_1}{1 + b_0}. \quad (3)$$

This is the same labor decomposition as the companion manuscript’s $\phi\eta + (1-\phi)\mu + r + b_A$ numerator, with a smaller, non-reused notation set. All quantities are defined once in Appendix A. If a group’s baseline mean is zero, use the hours identity instead of an undefined effort ratio.

The exception case share p and its baseline labor share w need not coincide. A small group of difficult projects may retain much of the original work. The relation follows from the covariance of routing and baseline effort; Appendix B gives the short proof. This is a property of selection, not evidence that the benchmark’s failing cases are the costly enterprise cases. Human timing data are needed to establish that relationship.

Table 12: One transparent sensitivity illustration, not fitted to DGF-Bench. Both rows set $w = 0.40$, $b_0 = 0$ and use the same case partition.

Assumed regime	a	u	r	b_1	R	FTE from 100
Low residual effort	0.25	0.05	0.02	0.05	0.20	20
High residual effort	1.50	0.50	0.10	0.10	1.10	110

The difference in the table is produced by the stated assumptions. It is not a discovered effect, a fitted forecast, or an estimate obtained from the model success rates. Its use is to show exactly what would need to be measured: assistance on exceptions, residual standard review, additional rework, and recurring support. The long manuscript’s multiple workforce scenarios are omitted here because repeating such arithmetic does not add empirical evidence.

5.3 Implications for staffing and employment

If validated review automation lowers L_1 at comparable output and quality, fewer workload-equivalent FTE are required for those activities. This is a mechanism through which work previously performed by employees can be replaced. A company may then reduce staffing, avoid new hires, redeploy people, or increase reviewed volume. Those are observable choices, not identical outcomes. Required hours, allocated staffing, headcount, and hiring should be reported separately so that the workforce claim can be evaluated rather than inferred from a technology demonstration.

An organization could reduce headcount while leaving unmet review demand or transferring unpaid work to another team. Conversely, it could preserve headcount while governing more projects with the same effort. The primary comparison therefore standardizes volume and case mix, retains quality and service requirements, and includes shared supplier support. Initial implementation effort should also be reported for the investment decision, even when the primary endpoint concerns recurring annual work.

The model results show that models can execute many represented tasks, while the deterministic control establishes that this scaffold does not require a model. Neither contains human hours or estimates R . A prospective 80% reduction in required DGF FTE is a hypothesis about a possible deployment outcome, not the numerical complement of model accuracy. Its new registration-anchored seven-year horizon is specified below; the original 2033 projection remains a separate historical claim.

6 A feasible validation program

The empirical gaps require additional observations. The following human and field studies are proposed; no human participants, enterprise cohort, or independent semantic adjudication have been

collected for this paper. Their protocols should be finalized, independently reviewed, and registered before new outcome collection. Existing benchmark inspections and design choices must be disclosed as prior information. The companion files in `paper2/protocols/` provide an operational starting point rather than a claim of preregistration.

6.1 Human and deterministic baselines

A fair baseline must first specify which ability is being compared. A human with only Word documents and a model with authoritative structured facts are not solving the same information problem. The first comparison should therefore give qualified reviewers and agents the same policy, accessible evidence, mandate information, output requirements, and case population. Humans can use a form that captures structured outputs; record form-entry time separately from investigation and deliberation. The exact-quotation requirement should be explained to both groups, while a separate semantic evaluation tests support independently of serialization.

Evaluate at least three conditions: unaided human review, agent review, and human review of an agent-produced draft. Retain the deterministic policy control now evaluated in Section 3.2, and compare it with the same future held-out cases. Its current result establishes sufficiency of the supplied rule scaffold; independent cases must test its generalization. Keep extraction from unstructured documents as a separate experimental condition. An oracle that reads the answer key can check the harness, but cannot serve as a substantive baseline.

Recruit domain-qualified participants, record relevant experience, and train them on development cases excluded from evaluation. Assign held-out cases with counterbalanced order so that a participant does not review the same case once unaided and again with an agent draft. Preserve route-level context when measuring complete routes. Independently adjudicate answer quality without revealing the review condition where practical. Report compensation, missing sessions, exclusions, and any identifiable information retained under the study’s ethics requirements.

Primary outcomes should include independently adjudicated disposition quality, false approval and critical-miss rates, semantic evidence support, and active human minutes. Total cycle time and exact-format conformity are secondary outcomes. For the assisted arm, count time verifying the draft, finding errors, rewriting outputs, and escalating issues; a short initial reading time alone is not an efficiency gain. Record time spent by other specialists who receive questions from the assigned reviewer.

The number of cases and reviewers should follow a prospective power and precision analysis using a stated noninferiority margin, expected paired disagreements, and participant/case clustering. Three hundred synthetic cases do not by themselves determine the sample size needed to establish rare-error safety or workforce effects. A small pilot may estimate variance and operational burden, but its outcomes must not be used repeatedly to choose a favorable primary metric. A confirmatory set should remain untouched until the analysis is frozen.

6.2 Independent cases, scaffold removal, and repeated trials

The generator and evaluator share an underlying rule system. An independent set should be written or adjudicated by specialists who did not construct those rules, using policies and documents whose ambiguities are documented. Specialists should be allowed to conclude that a case is underdetermined. Such cases test escalation rather than forcing a single convenient answer. Keep both policy disagreements and unambiguous implementation errors visible.

On a held-out synthetic set, compare full structured facts with a document-only condition, then introduce stale evidence, conflicting versions, and incomplete records using frozen transformations. Validate that the necessary facts remain recoverable when the task is intended to be solvable. A drop after hiding `REVIEW_FACTS` would quantify dependence on the scaffold; it would not by itself

distinguish extraction difficulty from reasoning difficulty. Add an extraction-only measurement so that these components can be examined separately.

Repeat trajectories for every selected model–case pair using a predeclared repeat count, reporting all attempts rather than selecting the best. Record provider, timestamps, resolved model identifiers, inference settings, resource limits, and retries. Separate sampling over projects from variability over repeated executions. If costs constrain replication, use a stratified case sample whose selection is fixed before outcomes and retain its inclusion probabilities. Expand the endpoint set to test whether the observed ranking survives a broader model selection; the current three endpoints are not a population sample of AI systems.

Adversarial document content and prompt injection should form a separately declared condition. Such tests concern whether the reviewer maintains its rules and authority while reading untrusted evidence, as distinct from applying a trusted snapshot correctly. A favorable score on the current dataset provides no estimate of that failure rate.

6.3 Enterprise deployment: quality and labor together

Field evaluation should begin with shadow review: the system produces outputs while existing authority remains unchanged. Independently compare the outputs and measure the human work needed to correct them. Shadow operation establishes review performance and integration burden; it does not by itself demonstrate savings because it may add a parallel review. A later controlled stage must measure actual substitution of defined activities under the approved operating design.

Where feasible, randomize comparable workflows or teams to staged adoption and retain a contemporaneous comparison group. If randomization is not feasible, register a design using pre-adoption trends, case-mix adjustment, and explicit confounders such as policy changes, reorganization, demand changes, and other automation. A before–after reduction alone establishes a temporal change, not attribution to the agent. Report both the volume-standardized effect and the actual workload trajectory.

The activity ledger should include evidence preparation, ordinary review, exception handling, remediation coordination, quality checks, incident work, and recurring system support. Shared platform teams and supplier operators must be counted once, then allocated under a frozen rule. Unobserved external effort is missing, not zero. Maintain an obligation register so that apparent savings cannot be achieved by leaving corrective work unfinished or pushing it beyond the measurement window. The observation period must include scheduled maintenance and policy updates.

Quality requirements should be registered with domain owners before deployment: admissible decision error, critical misses, unauthorized actions, evidence validity, and service delay. Noninferiority on an average score is insufficient if it hides unacceptable critical failures. The primary labor reduction is interpretable only while those requirements are met. If a site reverts to manual review, its recovery and manual effort remain part of the outcome.

6.4 A prospective, registration-anchored workforce hypothesis

For a new cohort, the proposed primary hypothesis is an 80% reduction in required DGF FTE at comparable output, quality, and service over seven years from the end of a prospectively measured baseline. Let t_0 be the registered baseline start, t_B its end twelve months later, and $t_F = t_B + 7$ calendar years. Compare mean monthly standardized labor in the baseline year with that in the twelve months ending t_F , retaining support in both measurements. The calendar dates, sampling frame, measurement boundary, and analysis must be fixed at registration before baseline collection. No cohort is currently registered, so this paper assigns no fictitious registration date or completed enrollment.

As an illustration only, a baseline beginning 1 October 2026 would end 30 September 2027 and its seven-year endpoint year would end 30 September 2034. This is a new prospective hypothesis, not a renamed test of the companion manuscript’s 2026–2033 prediction. That earlier prediction requires an auditable reconstruction of the twelve months ending 20 September 2026 and cannot be confirmed from a later baseline. We retain it as historical context rather than the primary projection in this empirical paper’s abstract.

At registration, specify organizational weighting, case-mix adjustment, supplier coverage, attrition, and uncertainty from time recording. Confirmation requires the upper bound of the registered uncertainty interval for the final/baseline ratio to be at most 0.20, together with all registered quality and service criteria passing. A lower bound above 0.20 contradicts the threshold for that population; an interval spanning it is inconclusive. Missing material labor prevents confirmation. If no eligible cohort is enrolled, the hypothesis remains untested. The endpoint must not be postponed after outcomes are observed.

A convenience cohort supports claims about its participants. Ecosystem-wide inference requires a defensible sampling frame covering functions and suppliers. Required human work, staffing, headcount, and hiring are distinct outcomes. Observing the labor threshold does not establish that AI caused it; causal attribution requires the comparative deployment design above. The prospective protocol makes a large reduction measurable without presenting a future enrollment as a completed study.

7 Conclusion

DGF-Bench shows that the tested models can complete many specified governance reviews under explicit rules and accessible structured facts. Gemini attains 94.98% strict gate success and 230 completely conforming routes among 299 evaluable projects. Its dispositions are correct throughout that evaluable sample; its strict failures concern evidence conformity. Luna and DeepSeek also show substantial task capability, with lower strict success and observed substantive errors. The released traces permit these outcomes to be inspected instead of reducing them to a single demonstration or a single accuracy number.

The executed rules control achieves complete strict success on all 300 dossiers without a language model. This strengthens the evidence that the scaffolded review process can be automated, while removing any inference that a language model is needed for this result. The incremental value of models for unstructured evidence or ambiguous policies is not established here. If comparable performance can be achieved on operational evidence with valid authority and acceptable assurance costs, automation could replace work currently performed by employees and reduce the staffing required for DGF activities. That is the task-substitution hypothesis advanced in this paper. An FDE contributes by making the policy, evidence, mandates, handoffs, recovery, and human-work measurement function as one operating system. The recurring work required to maintain it belongs in the result.

The deterministic control and structural evidence audit are now complete. The next decisive evidence is a comparison with qualified humans, independently adjudicated support, scaffold-removal tests, repeated model trajectories, and measured human hours in deployment. The proposed studies are designed to separate a formatting repair from better judgment, task automation from merely parallel review, and a staffing cut from lower required work. The new prospective 80% FTE hypothesis uses a seven-year horizon from the end of a registered baseline; no cohort has yet been enrolled. It cannot retrospectively confirm the companion manuscript’s original 2033 prediction. Complete governance automation is not established as a law. The substantive claim is ambitious and testable: reliable execution of review contracts may substantially reduce the human work required to govern projects.

Data, code, and research status

The [DGF-Bench repository](#) hosts the public materials. The release [dgf-bench-300-20260923](#) contains the full recorded run (17,700 original files), prepared dataset (27,947 files), and benchmark source snapshot (1,099 files). It includes Word documents, architecture diagrams, simulated records, policies, traces, scores, usage ledgers, and historical failed attempts. Checksums and per-file inventories preserve provenance. Evaluator-only references are released for audit; they were not exposed by agent tools during the run.

The final data tables and reproduction instructions are in [research/2026-09-dgf-bench/](#). Offline reproduction uses the frozen source archive, not an assumption that the evolving default branch is identical. The earlier 15-project pilot is separately available, including its Word documents, under [experiments/run_20260922_180730_063947/](#); its outcomes are not pooled into this study. The second manuscript sources are in [paper2/](#). Its inherited benchmark tables and figures retain the original measurements. This revision adds a 300-case deterministic control and an exhaustive structural audit of Gemini’s failed evidence gates, released in [research/2026-09-followup/](#). No human adjudication or enterprise enrollment has been completed. Paid repetitions are planned separately and are not presented as completed results.

The author is Jeremy Canale, contact@jeremycanale.com, <https://www.jeremycanale.com>, <https://www.linkedin.com/in/jcanale13>. The workflow specification is author-designed; no confidential enterprise documents or employment outcomes are represented as observations. AI assistance was used to prepare and revise this manuscript and its supporting software; this is not independent expert adjudication. The author remains responsible for checking the claims, artifacts, and interpretation before submission.

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A Notation and accounting boundary

Table 13: Workload notation. Each symbol has a single role within this manuscript.

Symbol	Definition
V	Reviews per period, held comparable between regimes.
h_0	Mean baseline case-handling hours.
p	Share of cases assigned to human exception handling.
h_E, h_S	Post-deployment human hours per exception and standard case.
h_W	Additional rework hours per review, averaged over all cases.
B_0, B_1	Case-independent support hours in baseline and deployment regimes.
L_0, L_1	Total human hours per period within the same boundary.
K	Fixed useful-hours convention per FTE per period.
N_0, N_1	Required workload-equivalent FTE; $N_j = L_j/K$.
w	Exception group’s share of baseline case-handling labor.
a, u	Post-deployment effort ratios against each group’s own baseline.
r, b_0, b_1	Rework and support normalized by baseline case-handling effort.
R	Required FTE ratio at comparable volume and quality.

The dimensional identity is valid at $p = 0$ or $p = 1$ using only the group that exists. Group-relative ratios require positive baseline group means. The constraint $0 \leq w \leq 1$ does not constrain a or u to be at most one: review or exception work can increase after deployment. Fixed support is not eliminated by multiplying every term by a volume adjustment. If volumes change, separate fixed support from variable review effort before standardization.

An activity should be counted exactly once regardless of departmental ownership or supplier boundary. For example, a specialist’s correction after an agent approval is either exception handling or additional rework under the declared coding rule, not both. A shared evaluation team’s hours are counted once before allocation across workflows. A repeat gate visit may be included in review volume or in per-project effort; the study must choose a consistent basis. Elapsed waiting time and active human time are distinct, although exclusive monitoring may consume human capacity. Record the rule used for that case.

B Residual selection: the short result

Proposition 1. *Let $T \geq 0$ be integrable baseline handling time with mean $h_0 > 0$. Let J indicate assignment to human exception handling, $p = \Pr(J = 1) > 0$, and $e(T) = \Pr(J = 1 \mid T)$. The exception group’s baseline labor share satisfies*

$$w = \frac{\mathbb{E}[TJ]}{h_0} = p + \frac{\text{Cov}(T, e(T))}{h_0}.$$

Consequently it exceeds the case share exactly when routing is positively associated with baseline effort. If e is nondecreasing, this covariance is nonnegative.

Proof. Conditional expectation gives $\mathbb{E}[TJ] = \mathbb{E}[Te(T)]$ and $\mathbb{E}[e(T)] = p$, giving the identity. For independent copies T, T' , $2 \text{Cov}(T, e(T)) = \mathbb{E}[(T - T')(e(T) - e(T'))]$. The integrand is nonnegative when e is nondecreasing. $\square$

No claim about the conditional variance of exception times follows from the proposition. Estimating w requires baseline hours under a frozen routing partition, not just a count of cases later failed by a model. Changes in the routing rule require a new partition or a documented reweighting. The identity supports measurement of concentration of remaining work; it does not predict its size without those measurements.

C Verification and follow-up record

The original strict outcomes remain the primary results of the existing experiment. A future semantic audit should retain item identifiers, observed evidence paths, cited claims, independent labels, disagreement reasons, adjudicated labels, and sampling probabilities. Release the rubric and analysis version with the results. Keep the lexical and semantic scores side by side so that differences can be traced to particular findings rather than only to a revised headline rate.

The field protocol must record registration identifier, enrollment date, eligible population, baseline provenance, operating boundary, quality thresholds, primary labor outcome, uncertainty procedure, and missing-data rules before outcome analysis. None of those identifiers is invented for this manuscript. If historical records cannot support the 2026 baseline, a new prospective study must state its own dates and hypothesis explicitly.

The repository’s `paper2/REVIEW_RESPONSE.md` maps the methodological criticisms to editorial changes and outstanding empirical work. The `paper2/verify_manuscript.py` utility checks the inherited tables, figure hashes, headline result counts, and the declared sensitivity illustration against the released sources. These checks concern internal consistency and provenance, not independent validation of the underlying enterprise rules.